FORMULATION AND EVALUATION OF IMMEDIATE-RELEASE TABLETS OF NEBIVOLOL HYDROCHLORIDE

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ABSTRACT

Nebivolol is a beta-1 receptor blocker with a nitric oxide-potentiating vasodilatory effect and is used in the treatment of hypertension and left ventricular failure. Because nebivolol is poorly soluble in water, the objective of the study was to formulate immediate-release nebivolol tablets using superdisintegrants and a solubilizer by wet granulation to improve disintegration and dissolution and thereby obtain a faster onset of action. Microcrystalline cellulose, crospovidone, and croscarmellose sodium were used in the formulations, with pregelatinized starch as the binder. The prepared tablets were evaluated for physicochemical characteristics including thickness, weight uniformity, drug content, disintegration, in vitro drug release, and stability. As reported in the source manuscript, formulation F6 showed 98% in vitro drug release at 60 min, drug content of 98.12%, a disintegration time of 20 s, and satisfactory stability.

Keywords: Nebivolol hydrochloride; immediate-release tablet; wet granulation; croscarmellose sodium; crospovidone; dissolution; stability.

1. INTRODUCTION

The oral route is widely used for conventional drug delivery because it offers convenience, ease of administration, flexibility in dosage-form design, ease of production, and relatively low cost. Tablets are compact unit dosage forms that can provide dose precision, content uniformity, acceptable stability, convenient packaging, and suitability for large-scale manufacture. The performance of an immediate-release tablet depends on appropriate formulation design and the rapid disintegration and dissolution of the active pharmaceutical ingredient.

Nebivolol is an antihypertensive drug with poor aqueous solubility. The present work investigated immediate-release nebivolol hydrochloride tablets prepared by wet granulation using different concentrations of superdisintegrants and other excipients. The objective was to compare the trial formulations on the basis of their physicochemical properties and dissolution performance and to identify an optimized formulation.

2. MATERIALS AND METHODS

2.1 Formulation design

Eleven formulations (F1-F11) were prepared. The source manuscript reports a target tablet weight of 240 mg. The formulation composition is reproduced below from the PDF source. The PDF table reports 21.76 mg of nebivolol in each formulation; elsewhere in the manufacturing description, the source states that each tablet contained 6 mg of nebivolol hydrochloride. This discrepancy is retained here and should be verified against the original experimental record before submission.

Ingredient (mg)	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Intragranular											
Nebivolol	21.76	21.76	21.76	21.76	21.76	21.76	21.76	21.76	21.76	21.76	21.76
MCC PH 101	135.24	134.04	133.14	135.24	134.04	133.14	145.14	137.94	133.14	131.88	133.88
Mannitol	30	30	30	30	30	30	30	30	30	30	30
SLS	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	-	0.6
CCS	-	-	-	2.4	3.6	4.8	4.8	4.8	4.8	5.76	3.84
CP	2.4	3.6	4.8	-	-	-	-	-	-	-	-
Binder solution											
Pregelatinized starch	24	24	24	24	24	24	12	19.2	24	24	24
Water	q.s.	q.s.	q.s.	q.s.	q.s.	q.s.	q.s.	q.s.	q.s.	q.s.	q.s.
Extragranular											
MCC PH 102	20	20	20	20	20	20	20	20	20	20	20
CCS	-	-	-	2.4	3.6	4.8	4.8	4.8	4.8	3.84	5.76
CP	2.4	3.6	4.8	-	-	-	-	-	-	-	-
Magnesium stearate	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Aerosil	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Total tablet weight	240	240	240	240	240	240	240	240	240	240	240

Table 1. Formula for preparation of nebivolol hydrochloride tablets (reproduced from the source PDF).

2.2 Preparation of tablets

The tablets were prepared by wet granulation. Intragranular ingredients were passed through a #40 mesh and mixed. Pregelatinized starch was dispersed in water to prepare the binder solution, and the dry blend was granulated to the desired end point. The wet mass was dried at 55 +/- 5 °C until the desired loss on drying was achieved. Dried granules were passed through a #30 mesh; retained material was milled through a 0.5-mm screen using an oscillating granulator. Extragranular ingredients were passed through a #40 mesh, whereas magnesium stearate was passed through a #60 mesh. The granules were blended, lubricated, and compressed using a rotary CADMACH tablet-compression machine.

2.3 Evaluation of granules

Bulk density: Bulk density was determined from the mass of the powder and its untapped bulk volume (BD = m/V0).

Tapped density: Tapped density was determined after tapping a known mass of powder in a graduated cylinder (TD = m/Vt).

Carr's compressibility index: Carr's index was calculated as [(tapped density - bulk density)/tapped density] x 100.

Hausner ratio: Hausner ratio was calculated as tapped density divided by bulk density.

Angle of repose: Angle of repose was determined from the height and radius of the powder cone using theta = tan^-1(h/r).

2.4 Evaluation of tablets

Prepared tablets were evaluated for general appearance, thickness, hardness, weight variation, friability, and drug content. Hardness was determined using a Monsanto hardness tester and the mean of five determinations was reported. Friability was measured using a Roche friabilator at 25 rpm for 100 rotations (4 min); the percentage friability was calculated as (W1 - W2)/W1 x 100. The source manuscript states an acceptance limit of 1.0%. Tablet thickness was measured with vernier calipers.

For drug-content determination, three tablets from each formulation were weighed individually and powdered. A quantity of powder equivalent to 10 mg was transferred to a 100-mL volumetric flask, diluted with 0.01 N HCl, suitably diluted, and analyzed spectrophotometrically at 269 nm against a blank.

2.5 In vitro dissolution study

Dissolution testing was performed in triplicate using USP Apparatus II (paddle) with 900 mL of 0.01 N HCl at 37 +/- 0.5 °C and 50 rpm. Samples (5 mL) were withdrawn at specified intervals, filtered through a 0.45-micrometer filter, and replaced with an equivalent volume of fresh dissolution medium. Samples were analyzed by UV spectrophotometry at 269 nm. The source PDF lists dissolution sampling times of 5, 10, 15, 20, 25, and 30 min in one section and presents graphical data extending to 45 and 60 min elsewhere; both are reported as present in the source.

2.6 Drug-release kinetics

The source manuscript describes model-dependent analysis of dissolution data using release models including zero-order and first-order kinetics. The supplied plots for the optimized formulation show first-order and zero-order fits.

2.7 Stability studies

The optimized formulation was subjected to stability studies at 25 +/- 2 °C/60 +/- 5% RH and 40 +/- 2 °C/75 +/- 5% RH for 3 months. Physical characteristics, drug content, and in vitro drug-release profiles were evaluated.

3. RESULTS AND DISCUSSION

3.1 Dissolution performance

The source manuscript identifies F6 as the optimized formulation and reports a maximum drug release of 98%. The abstract specifies 98% release at 60 min. The graphical comparisons supplied in the PDF show cumulative percentage drug release for F1-F6, F7-F11, a comparison between the innovator and F6, and the time course for F6.

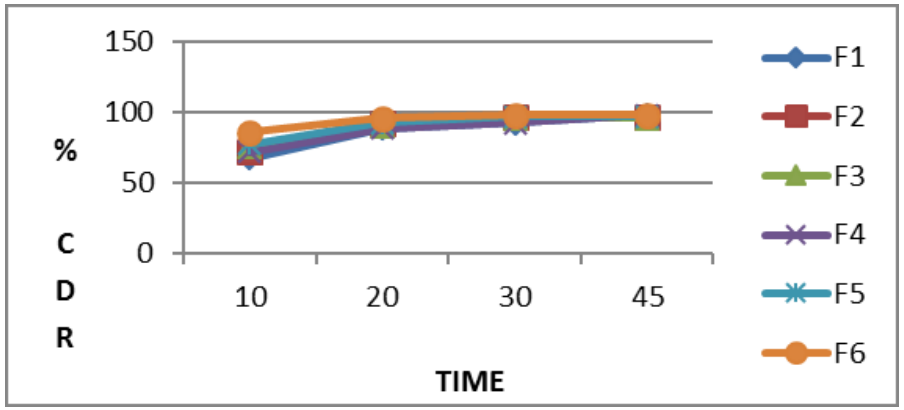


Figure 1. Cumulative percentage drug release for formulations F1-F6 (source PDF).

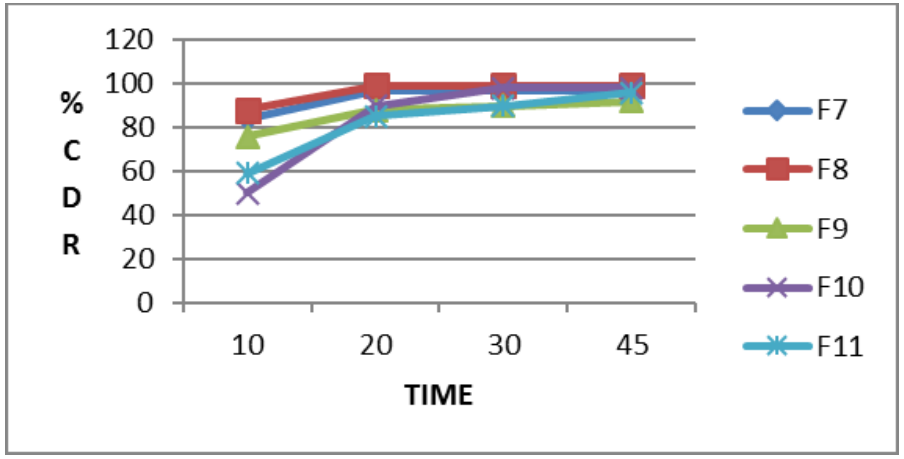


Figure 2. Cumulative percentage drug release for formulations F7-F11 (source PDF).

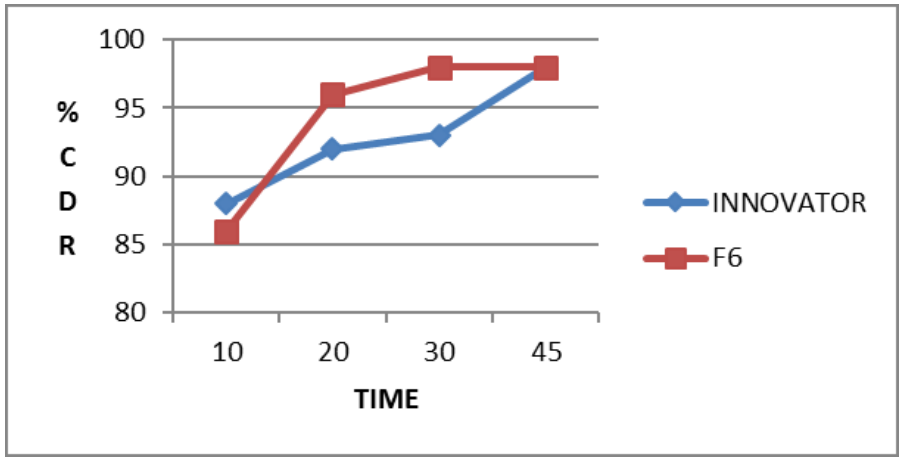


Figure 3. Comparative dissolution profile of the innovator and optimized formulation F6 (source PDF).

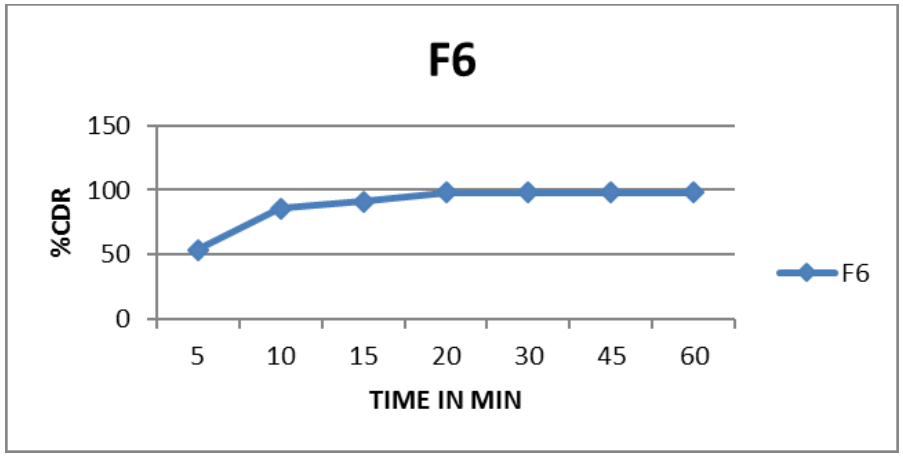


Figure 4. Cumulative percentage drug release profile for formulation F6 (source PDF).

3.2 Physical characterization

The source manuscript reports that tablet weight variation was within the prescribed Indian Pharmacopoeia limit of +/-7.5%. Tablet hardness ranged from 5.0 to 6.5 kg/cm². Friability values were reported to range from 0.65% to 1.8%. Because the same source states a friability limit of 1.0%, formulations at the upper end of the reported range would exceed that stated criterion. Drug content across formulations was reported to range from 98.68% to 101.1%.

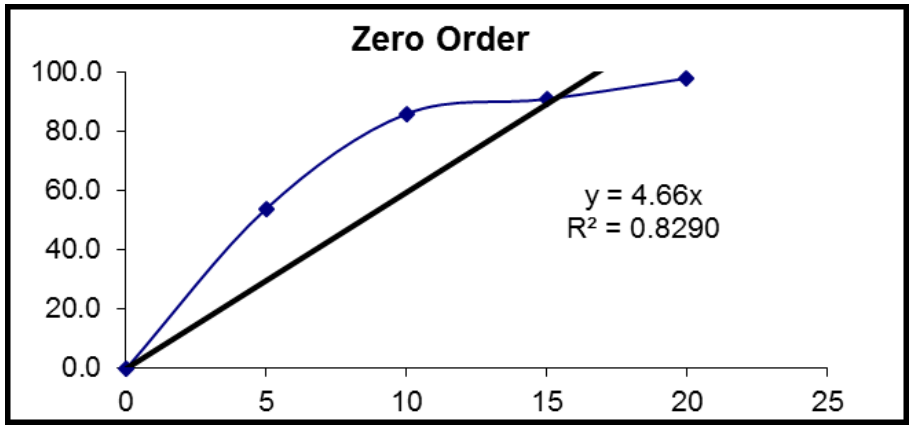


Figure 5. First-order release plot reported in the source PDF.

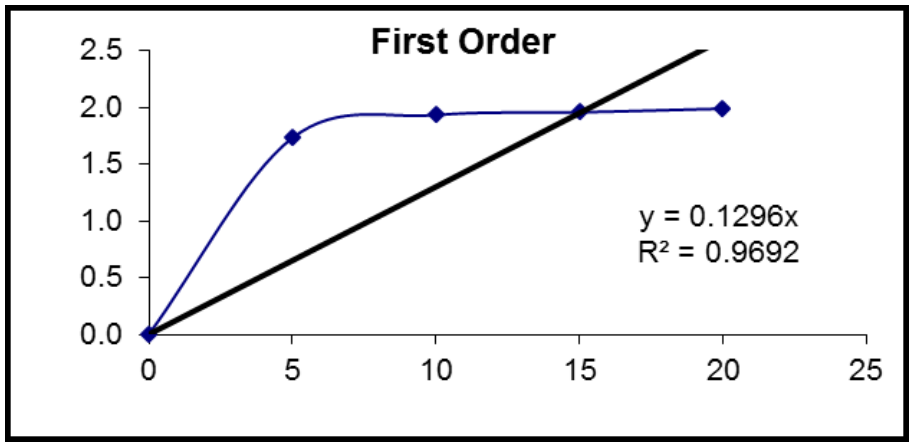


Figure 6. Zero-order release plot reported in the source PDF.

4. CONCLUSION

Immediate-release nebivolol hydrochloride tablets were prepared by wet granulation and evaluated for physicochemical and dissolution characteristics. Among the formulations investigated, F6 was identified in the source manuscript as the optimized formulation and was reported to provide 98% drug release, drug content of 98.12%, a disintegration time of 20 s, and satisfactory stability. The formulation and analytical discrepancies identified in the source PDF, particularly the stated nebivolol amount per tablet and the friability range relative to the stated acceptance criterion, should be reconciled with the original laboratory records before journal submission.

5. DECLARATIONS

Conflict of interest: The author declares no conflict of interest.

Funding: No external funding information was reported in the source manuscript.

Data availability: The data summarized in this manuscript are those reported in the supplied source document.

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